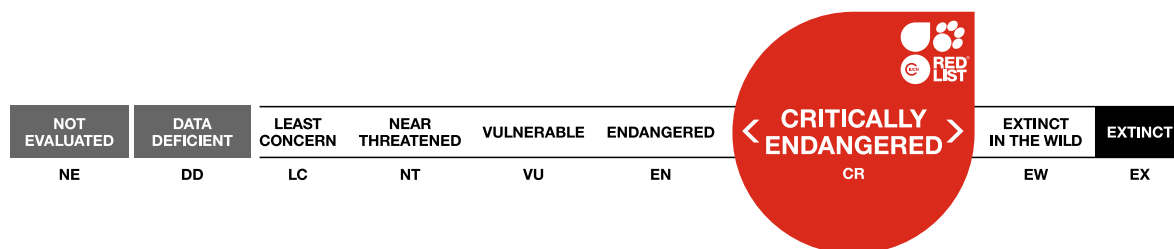


Pongo tapanuliensis, Tapanuli Orangutan

Amendment version

Assessment by: Nowak, M.G., Rianti, P., Wich, S.A., Meijaard, E. & Fredriksson, G.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Primates	Hominidae

Scientific Name: *Pongo tapanuliensis* Nurcahyo, Meijaard, Nowak, Fredriksson & Groves, 2017

Common Name(s):

- English: Tapanuli Orangutan
- French: Orang-outan de Tapanuli
- Spanish; Castilian: Orangután de Tapanuli
- German: Tapanuli-Orang-Utan

Taxonomic Source(s):

Nater, A., Greminger, M.P., Nurcahyo, A., Nowak, M.G., de Manuel Montero, M., Desai, T., Groves, C.P., Pybus, M., Sonay, T.B., Roos, C., Lameira, A.R., Wich, S.A., Askew, J., Davila-Ross, M., Fredriksson, G.M., de Valles, G., Casals, F., Prado-Martinez, J., Goossens, B., Verschoor, E.J., Warren, K.S., Singleton, I., Marques, D.A., Pamungkas, J., Perwitasari-Farajallah, D., Rianti, P., Tuuga, A., Gut, I.G., Gut, M., Orozco-terWengel, P., van Schaik, C.P., Bertranpetit, J., Anisimova, M., Scally, A., Marques-Bonet, T., Meijaard, E. and Krützen, M. 2017. Morphometric, behavioural, and genomic evidence for a new orangutan species. *Current Biology* 27: DOI: 10.1016/j.cub.2017.09.047.

Taxonomic Notes:

This species was first described in 2017 (Nater *et al.* 2017). The Nater *et al.* study showed that an isolated orangutan population in the Batang Toru region, which is the southernmost range of extant Sumatran Orangutans (*Pongo abelii*), south of Lake Toba, is distinct from other Sumatran and Bornean populations. Through a comparison of cranio-mandibular and dental characters from an orangutan killed during human-orangutan conflict to 33 adult male orangutans of similar developmental stage, the study found consistent differences between the Batang Toru individual and other extant Ponginae. Model-based approaches based on the analyses of 37 orangutan genomes supported the morphological results by revealing that the deepest split in the evolutionary history of extant orangutans occurred ~3.38 Ma ago between the Batang Toru population and those to the north of Lake Toba, while both currently recognized species separated much later at about 674 ka. The analyses show that there was a small gene flow between *P. abelii* and *P. tapanuliensis* until 10–20 ka. The combined analyses support a new classification of orangutans into three extant species.

Assessment Information

Red List Category & Criteria: Critically Endangered A4bcd [ver 3.1](#)

Year Published: 2024

Date Assessed: October 11, 2017

Justification:

Due to high levels of habitat conversion and fragmentation, and illegal killing, *Pongo tapanuliensis* is estimated to have experienced a significant population reduction in recent decades. Forest loss data

indicate that orangutan habitat below 500 m asl of both *P. tapanuliensis* and *P. abelii* was reduced by 60% between 1985 and 2007 (Wich *et al.* 2008, 2011). It is thought that this reduction will continue for the Tapanuli Orangutan as forests within its range remain under considerable threat (Wich *et al.* 2016). Significant areas of the Tapanuli Orangutan's range are seriously threatened by habitat conversion for small-scale agriculture, mining exploration and exploitation, a large-scale hydroelectric scheme, geothermal development, and agricultural plantations. Approximately 14% of the geographic range of the Tapanuli Orangutan is not protected, nor even allocated as forest estate, and even the protected areas are not immune from the above threats (Wich *et al.* 2008, 2011, 2016). The Tapanuli Orangutan is also hunted (Wich *et al.* 2012). Due to their slow life history, with a generation time of at least 25 years (Wich *et al.* 2004, 2009), orangutans on Sumatra are unable to sustain substantial and continual loss of individuals. A quantitative population viability analysis estimated that in 1985 there were ~1,489 individuals of the Tapanuli Orangutan, and that these would decline to only 257 over a 75-year period to 2060 (Sumatran Orangutan Conservation Programme unpublished data). With current management, the key threats of loss of lowland habitat and hunting are not being effectively reduced, and we therefore predict an 83% decline over the course of three generations. The species therefore qualifies as Critically Endangered under criterion A4.

Previously Published Red List Assessments

[2023 – Critically Endangered \(CR\)](#)

[2017 – Critically Endangered \(CR\)](#)

Geographic Range

Range Description:

For much of the 20th century, orangutans on Sumatra were thought to be restricted to the north and west of Lake Toba, in and around the Leuser Ecosystem. Populations south of Lake Toba were overlooked, even though a 1939 review of the species' range mentioned that orangutans had been reported in several forest areas in that region (Nederlandsch-Indische Vereeniging tot Natuurbescherming 1939). It was not until 1997 that these isolated orangutan populations were rediscovered (Meijaard 1997, Rijksen and Meijaard 1999). As a consequence of further sightings, published in 2003 (Wich *et al.* 2003), a study of the southern populations started in 2005 when the Sumatran Orangutan Conservation Programme (SOCP) began a conservation project in the mountainous Batang Toru region with the intent of conserving the orangutan population and their remaining habitat south of Lake Toba.

The only known population of *Pongo tapanuliensis* occurs in the uplands of the Batang Toru Ecosystem, an area of roughly 1,500 km² consisting of three forest blocks, of which 1,022 km² is suitable orangutan habitat (Wich *et al.* 2016). *Pongo tapanuliensis* is the least numerous of all great ape species.

Pongo tapanuliensis was until relatively recently more widespread, with sightings further south in the lowland peat swamp forests in the Lumut area (Wich *et al.* 2003) and several nests encountered during a rapid survey in 2010 (G. Fredriksson pers. obs.). The forests in the Lumut area have in recent years almost completely been converted to oil-palm plantations. In 2008, a captive Tapanuli Orangutan was confiscated in Lumut. Observation were also made of a male orangutan in the Dolok Ginjang area, just north of the Batang Toru West forest block in the Adiankoting subdistrict in North Tapanuli, during a

human conflict situation where an orangutan was shot at with an air rifle when it was found foraging on durian fruits (G. Fredriksson pers. obs.). Recent surveys in this area did not find any orangutan nests (Nowak *et al.* unpublished data).

A hunting trip report from 1879 described orangutan encounters near the small town of Mantinggi, south of the current population (0°58' N, 100°0' E) (Kramm 1879). In 1902, orangutan signs and reports were noted along several small rivers on the south side of the Sibolga Bay, west of Batang Toru (Miller 1903). Unsubstantiated reports suggest that orangutans, presumably *P. tapanuliensis*, may have occurred even further south in Sumatra in historic periods. A number of museum specimens are claimed to have come from Padang, Palembang and Jambi (Schlegel and Müller 1839–1844, Beccari 1904, Hooijer 1948, Veracini *et al.* 2010), and an orangutan kill was reported in 1930 near Palembang and along the Musi River toward the base of the Barisan Mountain range (Delmont 1930). These accounts suggest that isolated individuals or small populations of *P. tapanuliensis* may have survived outside the current range well into the 1900s, although without studying specimens from these areas, we cannot exclude that they could be *P. abelii*. Despite the distance to the south and their rarity, such reports are perhaps not surprising, as orangutans are long lived and isolated individuals or populations could potentially exist in fragmented landscapes for >50 years, provided that food resources were available and stochastic processes were rare.

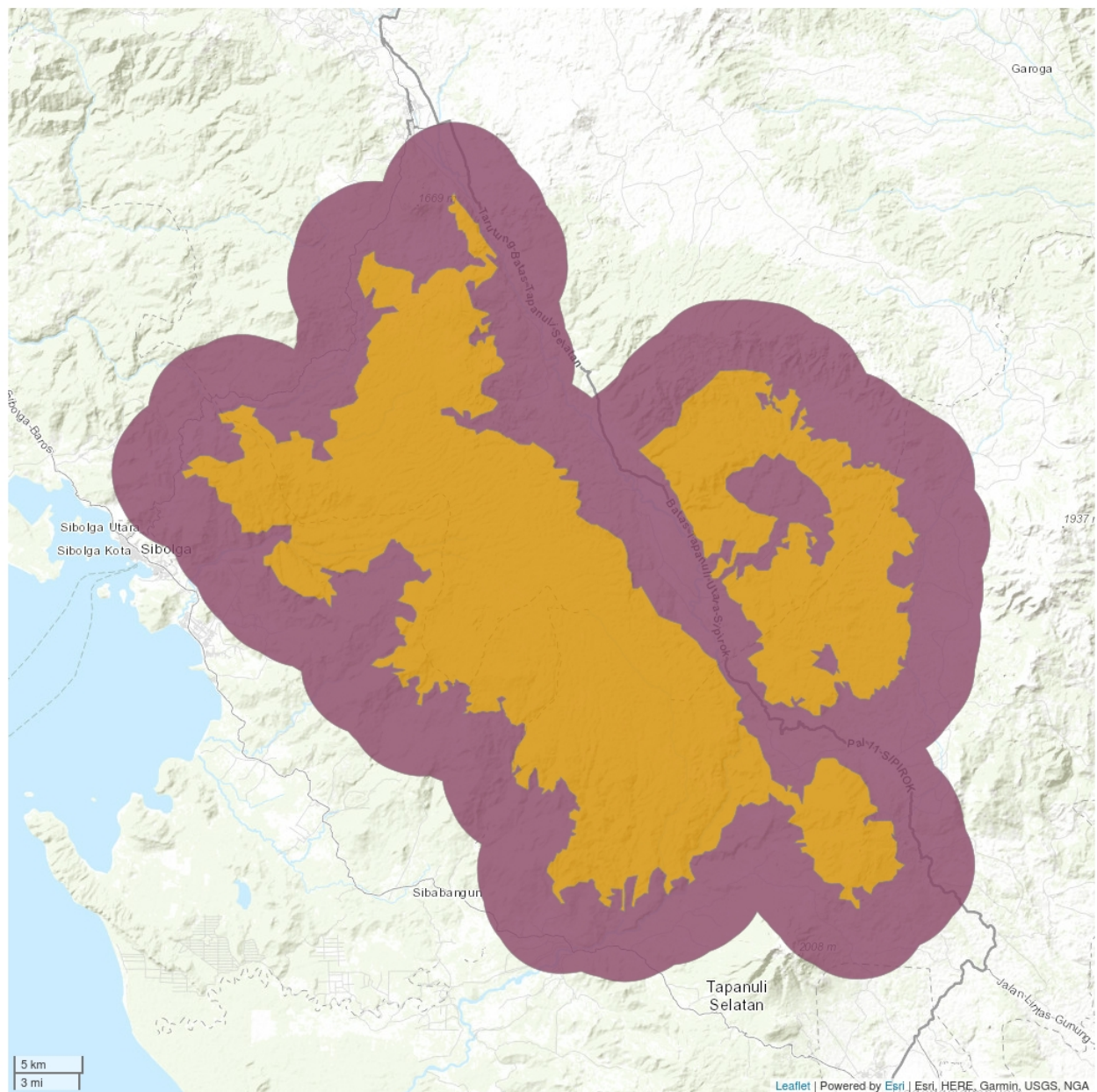
See the Supplementary Information for details of how the distribution map for this species was updated for the amended version of this assessment.

For further information about this species, see [Supplementary Material](#).

Country Occurrence:

Native, Extant (resident): Indonesia (Sumatera)

Distribution Map

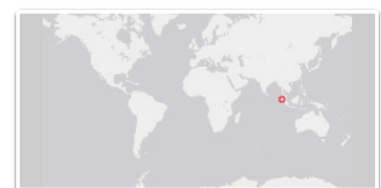
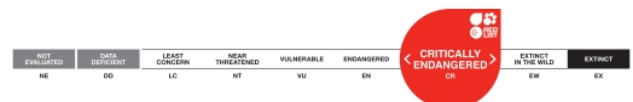


Legend

- EXTANT (RESIDENT)
- POSSIBLY EXTANT (RESIDENT)

Compiled by:

Wich, S.A., Ni'Matullah, S, Sherman, J., and Meijaard, E. 2023. Spatial data for *P. tapanuliensis* Red Listing. Borneo Futures, Brunei Darussalam 2023



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

With a population estimate of fewer than 800 individuals (Wich *et al.* 2016), *Pongo tapanuliensis* is the least numerous of all great ape species. Its distribution is separated by around 100 km from the closest population of *P. abelii* to the north. A combination of small population size and geographic isolation is of particularly high conservation concern, as it may lead to inbreeding depression (Hedrick and Kalinowski 2000) and threaten population persistence (Allendorf *et al.* 2013). Highlighting this, a recent study (Nater *et al.* 2017) discovered extensive runs of homozygosity in the genomes of two *P. tapanuliensis* individuals, pointing at the occurrence of recent inbreeding.

Only about 10% of the geographic range is in an area recognized by the World Database of Protected Areas. Another 76% is in Hutan Lindung (Protection Forest), and 14% does not have any 'forest status' in the spatial plans, despite the fact that it consists of rugged primary forest with the highest densities of Tapanuli Orangutans in the Batang Toru Ecosystem (SOCP unpublished data). This roughly translates to 8% of the total distribution located in nationally recognized conservation area (Cagar Alam, Strict Nature Reserve), 78% in Protection Forest (Hutan Lindung), and the remaining 14% of the range is entirely unprotected.

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

Despite a complex geological history, the underlying parent material within the geographic range of *Pongo tapanuliensis* consists primarily of recent (i.e., Quaternary, <2.588 Ma) igneous rock, most of which is acidic volcanic rock from the Toba super-eruption of ~74 ka (Subardja *et al.* 1990, Oppenheimer 2002, Barber *et al.* 2005). This is in stark contrast to the north Sumatran Orangutan population that persists in forests atop older (i.e., Tertiary and above, >2.588 Ma) igneous and sedimentary rock (Darul Sukma *et al.* 1990a, 1990b, 1990c, 1990d, Hidayat *et al.* 1990, Hikmatullah *et al.* 1990, Wahyunto *et al.* 1990, Barber *et al.* 2005). While comparative data in the format analysed for Sumatra are lacking for Borneo, previously published reconstructions of Borneo's geology highlighted a complex geological history that was in general different from that of Sumatra. This includes large swaths of landscape with pre-Cenozoic and Cenozoic sedimentary rock, with comparatively fewer areas characterized by igneous rock formations (Hall *et al.* 2008).

Using freely available WorldClim data (Hijmans *et al.* 2005), *P. tapanuliensis* can be differentiated from *P. abelii* by a highly narrow ecological niche, including a restricted upland elevation range (*P. abelii*: 701.7±454.8 m asl; *P. tapanuliensis*: 834.4±219.3 m asl), cooler average temperatures (*P. abelii*: 23.3±2.5°C; *P. tapanuliensis*: 22.1±1.2°C), and relatively high levels of average yearly rainfall (*P. abelii*: 2,435.1±460.5 mm/year; *P. tapanuliensis*: 2,607.5±231.2 mm/year). Similarly, compared to *P. pygmaeus*, *P. tapanuliensis* occurs at much higher elevation (*P. pygmaeus*: 170.6±187.0 m asl; *P. tapanuliensis*: 834.4±219.3 m asl), and experiences far lower average temperatures (*P. pygmaeus*: 26.0±1.0°C; *P. tapanuliensis*: 22.1±1.2°C), and lower levels of average rainfall (*P. pygmaeus*: 2,827.3±420.9 mm/year; *P. tapanuliensis*: 2,607.5±231.2 mm/year). At SOCP's long-term monitoring station, average monthly temperature from 2009–2015 was recorded as 22.0°C (±0.89°C, n=83 months), with a monthly range of 20.5–24.2°C, whereas average total rainfall from 2009–2015 was recorded as 3,654 mm per year (±390.0 mm, n=7 years), with a yearly range of 3,186–4,210 mm (SOCP unpublished data).

Provided that climate, parent material, and time are key factors in soil formation/quality, and thus forest productivity/quality (Tan 2008, Marshall *et al.* 2009, Wich *et al.* 2011, Bockheim *et al.* 2014), the aforementioned factors are key distinctions between the *P. tapanuliensis* and most Bornean and north Sumatran Orangutan populations. A recent study showed that forest compositions experienced by *P. tapanuliensis* are at present mainly known from upland areas of southern Sumatra and Borneo, areas that notably do not currently maintain orangutan populations (Slik *et al.* 2011). They are largely restricted to primary forest, though they do venture into disturbed habitat at forest edges.

As noted by Wich and colleagues (Wich *et al.* 2014), *P. tapanuliensis* was observed feeding on a number of tree species that had not previously been recorded as orangutan food species. These unique species include *Agathis borneensis* from the Araucariaceae family, *Gymnostoma sumatranum* from the Casuarinaceae family, and *Dacrycarpus imbricatus*, *Dacrydium beccarii*, *Dacrydium comosum*, and *Podocarpus neriifolius* from the Podocarpaceae family. To our knowledge, occasional feeding on species from these families had only been documented in Gunung Palung, Borneo (fruits of an unidentified *Agathis* species and vegetation of an unknown *Casuarina* species [Casuarinaceae]), and Ketambe, Sumatra (bark of *Sundacarpus amara* [Podocarpaceae]) (Russon *et al.* 2009). In contrast, the five aforementioned conifer species (Araucariaceae and Podocarpaceae) and one non-conifer evergreen species (Casuarinaceae) represent 21.9% of all feeding observations (n=58,558 two-minute instantaneous samples) recorded between 2011 and 2015 at SOCP's long-term monitoring station in the Batang Toru Ecosystem (SOCP unpublished data). Thus, a significant proportion of the dietary profile of Tapanuli Orangutans is markedly different from that of previously studied orangutan populations.

Systems: Terrestrial

Use and Trade (see Appendix for additional information)

It is illegal to capture, injure, kill, own, keep, transport, or trade an orangutan.

Threats (see Appendix for additional information)

Due to the extremely rugged terrain, external threats have been primarily limited to illegal clearing of protected forests, hunting and killing during crop conflict, and trade in young orangutans (Wich *et al.* 2012, Wich *et al.* 2016). Encroachment and hunting have increased in recent decades, due to an influx of migrants from Nias Island, west of Sumatra, who settle on protected forest land on Batang Toru's forest edge where no land claims exist at present (Wich *et al.* 2012). In addition, despite land status changes from production forest to that of protection forest in 2014 (Ministry of Forestry of the Republic of Indonesia 2014), one company still maintains a controversial 300 km² logging permit located in primary forest within the current range of *Pongo tapanuliensis*. In the southwest corner of the Batang Toru Forest Complex a gold and silver mine is located, which has recently converted *ca* 3 km² of *P. tapanuliensis* habitat and retains controversial mining permits overlapping parts of the remaining *P. tapanuliensis* range. Uncontested land speculation related to the company's exploration is further threatening the primary forest. More recently, a hydro-electric development has been proposed in the area of highest orangutan density, which could impact roughly 100 km² of *P. tapanuliensis* habitat, or nearly 10% of the entire species population. It could also jeopardize the chances of maintaining habitat corridors between the western and eastern *P. tapanuliensis* ranges, and with two smaller strict nature reserves, which could also maintain small populations of *P. tapanuliensis*.

Conservation Actions (see Appendix for additional information)

To ensure the long-term survival of *Pongo tapanuliensis*, conservation measures need to be implemented swiftly. Due to the rugged terrain, external threats have been primarily limited to illegal clearing of forests, hunting, killings during crop conflict and trade in orangutans (Wich *et al.* 2012, Wich *et al.* 2016). A hydroelectric development has been proposed recently in the area of highest orangutan density, which could impact nearly 10% of *P. tapanuliensis*' population. This project might lead to further genetic impoverishment and inbreeding, as it would jeopardize chances of maintaining habitat corridors between the western and eastern range, and smaller nature reserves, all of which maintain small populations of *P. tapanuliensis*.

The Tapanuli Orangutan is also protected by international legislation by default because it is included within the old concept of *P. abelii* which is listed on CITES Appendix I.

Credits

Assessor(s): Nowak, M.G., Rianti, P., Wich, S.A., Meijaard, E. & Fredriksson, G.

Reviewer(s): Williamson, E.A. & Mittermeier, R.A.

Authority/Authorities: IUCN SSC Primate Specialist Group

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External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.6. Forest - Subtropical/Tropical Moist Lowland	Resident	Suitable	Yes
1. Forest -> 1.9. Forest - Subtropical/Tropical Moist Montane	Resident	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity
1. Residential & commercial development -> 1.1. Housing & urban areas	Ongoing	Minority (<50%)	Slow, significant declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
1. Residential & commercial development -> 1.3. Tourism & recreation areas	Ongoing	Minority (<50%)	Slow, significant declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.2. Small-holder farming	Ongoing	Minority (<50%)	Rapid declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.3. Agro-industry farming	Ongoing	Minority (<50%)	Very rapid declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
3. Energy production & mining -> 3.2. Mining & quarrying	Ongoing	Minority (<50%)	Rapid declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance	
4. Transportation & service corridors -> 4.1. Roads & railroads	Ongoing	Majority (50-90%)	Very rapid declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target)	Ongoing	Minority (<50%)	Very rapid declines
	Stresses:	2. Species Stresses -> 2.1. Species mortality	
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.2. Unintentional effects (species is not the target)	Ongoing	Minority (<50%)	Rapid declines
	Stresses:	2. Species Stresses -> 2.1. Species mortality	

5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.3. Persecution/control	Ongoing	Minority (<50%)	Rapid declines
Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.4. Unintentional effects: (large scale) [harvest]	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance		
7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.1. Increase in fire frequency/intensity	Ongoing	Minority (<50%)	Very rapid declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance		
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Ongoing	Whole (>90%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.3. Indirect ecosystem effects 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place research and monitoring
Action Recovery Plan: Yes
Systematic monitoring scheme: No
In-place land/water protection
Conservation sites identified: Yes, over entire range
Percentage of population protected by PAs: 1-10
Area based regional management plan: Yes
Occurs in at least one protected area: Yes
Invasive species control or prevention: Not Applicable
In-place species management
Harvest management plan: No
Successfully reintroduced or introduced benignly: Yes
In-place education

Conservation Action in Place
Subject to recent education and awareness programmes: Yes
Included in international legislation: Yes
Subject to any international management / trade controls: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action Needed	Notes
1. Land/water protection -> 1.1. Site/area protection	Ensure protection management of batang toru forests and obtain protected status for the 14% of forest habitat that is not yet legally protected
1. Land/water protection -> 1.2. Resource & habitat protection	Establish corridors and protect the entire species range
2. Land/water management -> 2.1. Site/area management	Improved management and law enforcement in all protected areas, non-protected forests and concessions
2. Land/water management -> 2.3. Habitat & natural process restoration	Habitats where concessions or roads are shown to be illegal and prosecutions taken place to be confiscated (or cancelled) and restored to former condition
3. Species management -> 3.2. Species recovery	Key corridors to be identified and established to maintain all populations that reach minimum genetically viable levels
5. Law & policy -> 5.1. Legislation -> 5.1.2. National level	-
5. Law & policy -> 5.1. Legislation -> 5.1.3. Sub-national level	-
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.2. National level	-

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed	Notes
1. Research -> 1.1. Taxonomy	Continue research on this new species
1. Research -> 1.2. Population size, distribution & trends	Continue surveys to monitor the population and continue to survey other forested areas to determine whether small populations occur elsewhere
1. Research -> 1.5. Threats	Chain of custody studies to publicise companies still engaged in (directly or supplied by) destruction of the batang toru area
2. Conservation Planning -> 2.2. Area-based Management Plan	Effective management plans needed for batang toru, fully endorsed by national, provincial and district governments
3. Monitoring -> 3.1. Population trends	Always needed
3. Monitoring -> 3.4. Habitat trends	Always needed

Additional Data Fields

Population
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: Yes
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Generation Length (years): 25
Movement patterns: Not a Migrant

Amendment

Amendment reason: This second amended version of the 2017 assessment was created to attach supplementary information. A previous amended assessment updated the distribution map.

The IUCN Red List Partnership



The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#).

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